Oguz Altan

M.Sc. in Electrical Engineering

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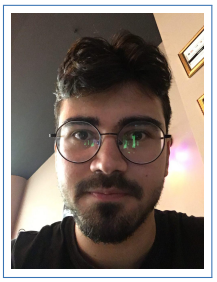

About

M.Sc. graduate from RWTH Aachen University in Electrical Engineering, Information Technology and Computer Engineering Department. Interested in intelligent systems, multi-agent systems, reinforcement learning, robotics, machine learning, and control theory. Research and application-oriented experiences supported by academic background and internships.

Education

2020–2023 **RWTH Aachen University**, *Aachen, Germany*, M.Sc, Electrical Engineering, Information Technology and Computer Engineering, Major: System and Automation, *GPA: 2.2/1.0*

February 2023–September 2023 **Fraunhofer FKIE**, *Wachtberg, Germany*, Master's Thesis,
Title: Tracking and Evasion using Co-Training with Context Knowledge, Grade: 1.3/1.0
-Multi-agent deep reinforcement learning
-Procedural generation
-Image processing with convolutional neural networks
-Path planning, tracking and evasion

September 2022 **Chalmers University of Technology**, *Gothenburg, Sweden*, Summer School, Analysis and Modelling Road User Behaviour

2016–2020 **Bilkent University**, *Ankara, Turkey*, B.Sc, Electrical and Electronics Engineering, *GPA: 3.35/4.00*

2011–2016 **Lycée de Galatasaray - Galatasaray High School**, *Istanbul, Turkey*, Science, *GPA: 86.20/100.00*

Work Experience

March–December 2022 **Siemens AG**, *Munich, Germany*, Engineering Intern and Working Student,
Artificial Intelligence on Wire Arc Additive Manufacturing (WAAMAI) Data Analytics Team
-Anomaly detection using various machine learning algorithms and frameworks
-Process monitoring and automation software development
-Edge computing with NVIDIA Jetson working on Ubuntu
-Closed loop controller optimization

June–September 2019 **Fraunhofer IIS**, *Erlangen, Germany*, Engineering Intern,
LIKE Department–PEMSY Team
-Redesign and optimization of PCB for IoT and Embedded Electronics team affiliated with FAU
-Microcontroller programming

June–July 2018 **TÜBİTAK Space Technologies Research Institute**, *Ankara, Turkey*, Engineering Intern,
Satellite Platform and Payload Electronic Design
-Signal processing and microprocessors

Notable Courses

B.Sc and M.Sc

Introduction to AI, Neural Networks, Reinforcement Learning and Learning-based Control, Data Science, Big Data Analytics, Robotics and Man-Machine-Interaction I and II, Simulation of Robotic Systems Sensors and Environment, Feedback Control Systems, Nonlinear Systems, Robust Feedback Theory, Digital Image Processing, Machine Learning for Speech and Audio Processing, Optimization in Engineering, Modeling of Mechatronic Systems, Laboratory: Audio Processing using Python, Seminar: Current Concepts and Trends in Robotics and Simulation Science

Projects

2021 **Mobile Robotics in Disaster Scenarios**, *Seminar: Current Concepts and Trends in Robotics and Simulation Science*, RWTH Aachen University

Review article submitted to the Institute of Man-Machine Interaction (MMI). Chapters: History of mobile robotics, techniques and systems developed for disaster scenarios, examples of mobile robots in disasters, the future of mobile robotics in disaster scenarios. For more, click here.

2020 **Accompanying Humans and Achieving Designated Tasks with Autonomous Mobile Robots (TAG-A-LONG-BOT)**, *Senior Year Project*, Bilkent University

Teamwork and collaboration of a group of five students in the Electrical and Electronics Engineering department, supported by the company ROKETSAN. Aims to design and implement autonomous mobile land robots to track a designated moving human, to avoid nearby obstacles by detecting and dodging them, and to smoothly move on the terrain via odometer information provided by beacons placed around the implementation zone. Main components are the *computer vision system*, *mechanical setup of the robot*, *control system*, *object tracking*, *avoidance simulations*, and *real-life testing*. For more, click here.

2018 **Hand Gesture Controlled Remote Car**, *Course Project: Microprocessors*, Bilkent University

4WD remote car controlled with the gesture of user's hand, communicating via Bluetooth. The car and the glove have microcontrollers: NXP FRDM-KL25Z and Arduino Nano, respectively.

2017 **Rotating Object Detector**, *Course Project: Introduction to Digital Design*, Bilkent University

Mechanism based on BASYS 3 FPGA Board programmed with VHDL that detects objects within a range and alerts the user. For demo, click here.

Skills

Programming Languages, *Python, MATLAB, Simulink, Java, LaTeX, Assembly, VHDL*

Technical Skills, *Linux, Version Control, VS Code, Docker, EAGLE*

Languages, *English (Advanced), French (Advanced), German (Beginner), Italian (Beginner), Turkish (Native)*

Honors and Awards

2022 **DAAD Scholarship for Completing Studies for Students**, DAAD and RWTH Aachen University

2016—2020 **Scholarship of the Turkish Prime Ministry**

2016 **Ranked 46th in Nationwide University Entrance Exam (LYS)**, Foreign Languages Exam

Examination Results

February 2020 **TOEFL iBT**

Reading 28/30 - Listening 22/30 - Speaking 23/30 - Writing 24/30 – 97/120 overall

February 2020 **GRE General Test**

Quantitative 166/170 - Verbal Reasoning 151/170 - Analytical Writing 4.0/6.0

Extracurricular Activities

2019–2020 **IEEE Bilkent Student Branch Vice Chair**

Administrating and supervising technical activities, lectures, conferences and competitions.

2018–2019 **IEEE Bilkent Student Branch Robotics and Automation Society (RAS) Coordinator**

Teaching fundamentals of electronics and Arduino microcontroller programming with C to preparatory and freshmen year students. Organizing technical conferences.

15 November 2023
Aachen, Germany